



Series €FGHE/C



Set-5

प्रश्न-पत्र कोड
Q.P. Code **55(B)**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।
Candidates must write the Q.P. Code on the title page of the answer-book.

भौतिक विज्ञान (सैद्धान्तिक)
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

PHYSICS (Theory)

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 70

Maximum Marks : 70

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 27 हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 35 प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains 27 printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 35 questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) *This question paper contains **35** questions. **All** questions are **compulsory**.*
- (ii) *This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*
- (iii) *In **Section A** – Questions no. **1** to **18** are Multiple Choice (MCQ) type questions, carrying **1** mark each.*
- (iv) *In **Section B** – Questions no. **19** to **25** are Very Short Answer (VSA) type questions, carrying **2** marks each.*
- (v) *In **Section C** – Questions no. **26** to **30** are Short Answer (SA) type questions, carrying **3** marks each.*
- (vi) *In **Section D** – Questions no. **31** to **33** are Long Answer (LA) type questions carrying **5** marks each.*
- (vii) *In **Section E** – Questions no. **34** and **35** are case-based questions carrying **4** marks each.*
- (viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 3 questions in Section D and 2 questions in Section E.*
- (ix) *Use of calculators is **not** allowed.*

Use the following values of physical constants, if required :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (} m_e \text{)} = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$



SECTION A

1. The electric flux through a Gaussian spherical surface enclosing a point charge q is ϕ . If the charge is replaced by an electric dipole, magnitude of its dipole moment being $2qa$, the flux through the surface will be :
(a) 2ϕ (b) ϕ
(c) $\frac{\phi}{2}$ (d) Zero
2. 10^9 electrons are transferred to a pith ball with charge 0.16 nC . Its charge now is :
(a) Zero
(b) $-3.2 \times 10^{-10} \text{ C}$
(c) $-1.6 \times 10^{-9} \text{ C}$
(d) $3.2 \times 10^{-10} \text{ C}$
3. The amount of charge flowing across a cross-section of a conductor, connected to a battery, in 4.0 s is 720 mC . The current in the circuit is :
(a) 0.18 A (b) 1.8 A
(c) 18 mA (d) 18 A
4. Two cells, one of emf $2E$ and internal resistance r and the other of emf E and internal resistance $r/2$ are connected in parallel, by connecting their positive terminals together and their negative terminals together. The equivalent emf of the combination is :
(a) $3E$ (b) $\frac{4E}{3}$
(c) $2E$ (d) $\frac{3E}{4}$



5. Two long straight parallel conductors carrying steady currents I_1 and I_2 in the same direction are separated by a distance d . The force exerted by one conductor on unit length of the other is :

- (a) $\frac{\mu_0 I_1 I_2}{4\pi d}$, attractive
(b) $\frac{\mu_0 I_1 I_2}{4\pi d}$, repulsive
(c) $\frac{\mu_0 I_1 I_2}{2\pi d}$, attractive
(d) $\frac{\mu_0 I_1 I_2}{2\pi d}$, repulsive

6. A long solenoid of length L and number of turns N carrying a current I produces a magnetic field B along its axis. If the current in the solenoid is doubled and its number of turns halved, the magnetic field along its axis will be :

- (a) $2B$ (b) $4B$
(c) $\frac{B}{2}$ (d) B

7. The speed of electromagnetic waves in vacuum is :

- (a) $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$ (b) $\sqrt{\mu_0 \epsilon_0}$
(c) $\mu_0 \epsilon_0$ (d) $\frac{1}{\mu_0 \epsilon_0}$

8. A thin concave lens of focal length $2f$ is put in contact with a thin convex lens of focal length ' f '. The focal length of the combination will be :

- (a) $2f$ (b) $\frac{f}{2}$
(c) $-f$ (d) f



9. In Young's double-slit experiment, the slit separation is made three times and the distance of the screen from the slits is doubled. The fringe width becomes :
- (a) 6 times (b) $\frac{3}{2}$ times
(c) $\frac{1}{6}$ times (d) $\frac{2}{3}$ times
10. The work function of potassium is 2.3 eV and threshold wavelength for it is 540 nm. The threshold wavelength for a metal whose work function is 4.6 eV will be :
- (a) 135 nm (b) 270 nm
(c) 540 nm (d) 1080 nm
11. The minimum energy required to free the electron from the ground state of the hydrogen atom in Bohr model is 13.6 eV. The energy of the electron in its second orbit will be :
- (a) - 6.8 eV (b) 6.8 eV
(c) - 3.4 eV (d) 3.4 eV
12. Which of the following series in hydrogen spectrum lies in its visible region ?
- (a) Lyman series
(b) Balmer series
(c) Paschen series
(d) Pfund series
13. Which of the following dopants in silicon will make it a p-type semiconductor ?
- (a) As (b) Al
(c) Sb (d) P



14. Which of the following has a negative temperature coefficient of resistivity ?
- (a) Silver (b) Copper
(c) Nichrome (d) Germanium
15. When a p-n junction diode is forward biased :
- (a) the depletion layer width increases and the barrier height is reduced.
(b) the depletion layer width decreases and the barrier height is reduced.
(c) both the depletion layer width and the barrier height increase.
(d) the depletion layer width decreases and the barrier height increases.

Questions number 16 to 18 are Assertion (A) and Reason (R) type questions. Two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false and Reason (R) is also false.
16. *Assertion (A) :* The potential energy of a magnetic dipole in a uniform magnetic field is maximum when its magnetic moment and the magnetic field are antiparallel to each other.

Reason (R) : A current loop behaves as a magnetic dipole.



17. *Assertion (A) :* The electrostatic field $\vec{E}$ is a conservative field.

Reason (R) : Line integral of electric field $\vec{E}$ around a closed path is non zero.

18. *Assertion (A) :* The photoelectric current increases with increase in intensity of incident radiation, keeping the frequency of the incident radiation and the accelerating potential fixed.

Reason (R) : The number of photoelectrons emitted per second is directly proportional to the intensity of incident radiation.

SECTION B

19. A closely wound coil having 100 turns carries a current of 2 A. If the magnetic flux linked with coil is $\pi^2 \times 10^{-5}$ Wb, find the radius of the coil. 2

20. (a) Identify the part of electromagnetic spectrum which is : 2
(i) suitable for radar systems.
(ii) sometimes referred to as 'heat waves'.

Write their wavelength range.

OR

(b) Write two characteristics of electromagnetic waves. Name the radiation used to kill germs in water purifiers. Write the range of their frequency. 2



21. (a) The intensities of the principal maximum and secondary maxima in a single slit diffraction pattern are not equal. Explain. 2
- OR**
- (b) In a Young's double-slit experiment, set-up in air, the fringe width is found to be β . If the entire set-up is immersed in water, how would β be affected ? Give reason for your answer. 2
22. An equiconvex lens of focal length f is cut into two equal halves perpendicular to its principal axis. Find the focal length of each half of the lens. 2
23. A nucleus with $A = 280$ splits into two nuclei X_1 and X_2 whose mass numbers are in the ratio $27 : 8$. Find the ratio of their radii, $\left(\frac{r_1}{r_2}\right)$ 2
24. Briefly explain how the valence band and conduction band are formed in a crystal. 2
25. Explain how a depletion region is formed in a p-n junction. 2

SECTION C

26. A cylindrical conductor of resistance R is connected to a battery. The drift velocity of electrons in this conductor is v_d . The conductor is disconnected from the battery and gradually stretched so that its length increases by 25%. It is again connected to the same battery. Find the new value of : 3
- (a) the drift speed of electrons, and
- (b) the resistance of the conductor.



- 27.** (a) A metallic rod of length 40 cm is rotated with a frequency of 300 rpm about a vertical axis passing through its centre. If a uniform magnetic field of 0.2 T acts vertically downward, find the emf induced between (i) the ends of the rod, and (ii) the centre and one end of the rod. 3

OR

- (b) A square coil of side 10 cm having 100 turns is suspended in a magnetic field of 100 mT so that its plane is normal to the field. The coil is rotated through 90° in 0.2 s. Calculate the average emf induced in it. 3
- 28.** State Lenz's law. Show that it is consistent with the principle of conservation of energy. 3
- 29.** (a) Obtain the relationship between atomic mass unit (u) and electron volt (eV).
(b) The mass of a ball is 0.5 kg. It is totally converted into energy. Calculate the energy output in eV. 3
- 30.** (a) Write Einstein's photoelectric equation. Use it to :
(i) explain the existence of threshold frequency for a given photosensitive surface.
(ii) show that the stopping potential, V_0 varies linearly with the frequency ν of incident radiation.

Explain how the value of Planck's constant can be found from the V_0 versus ν curve. 3

OR



- (b) An electron (charge $-e$, mass m) is revolving around a positively charged (charge $+e$) nucleus in a circular orbit of radius r .

3

- (i) For a stable orbit, equating the electrostatic Coulomb force between the electron and the nucleus with the requisite centripetal force, obtain an expression for radius r .
- (ii) What is Bohr's second postulate about angular momentum of electron in its orbit ? Using it and expression obtained in (i) above, show that the radius r of the n^{th} possible orbit is directly proportional to n^2 .

SECTION D

- 31.** (a) (i) Why do we use a small test charge to measure an electric field ?
- (ii) A small stationary positively charged particle is free to move in an electric field. In which direction will it begin to move ?
- (iii) Two point charges Q_1 ($40 \mu\text{C}$) and Q_2 ($-16 \mu\text{C}$) are placed along x-axis at 0 cm and 24 cm from the origin respectively. Calculate the net force on a third charge Q_3 ($-2.5 \mu\text{C}$) placed at $x = 36$ cm from the origin.

5

OR



(b) (i) How will the capacitance of a parallel plate capacitor change if :

(1) the plates area is doubled ?

(2) the separation between the plates is doubled ?

(ii) The effective capacitance of three capacitors of the same capacitance connected in series is $1 \mu\text{F}$. Find the :

5

(1) effective capacitance if they are connected in parallel.

(2) ratio of energy stored in the parallel combination of the capacitors to that in the series combination, if the combinations are connected to the same source one by one.

32. (a) (i) A charged particle with speed v_0 enters a region in which a strong and non-uniform magnetic field exists everywhere. It comes out of the region following a complicated trajectory without suffering any collision in the region. Would its final speed v be equal to its initial speed v_0 ? Justify your answer.

(ii) In a region, a uniform magnetic field of $6 \times 10^{-4} \text{ T}$ is maintained. An electron enters the field with a speed of $3 \times 10^6 \text{ ms}^{-1}$ normal to the field. Find radius of its path. Also calculate its energy in eV.

5

OR



- (b) (i) Derive an expression for the force acting on a conductor carrying current I in a magnetic field $\vec{B}$.
- (ii) A long horizontal conductor carries a current of 30 A, in east to west direction. What are the magnitude and direction of the magnetic field due to the current 1.0 m below the conductor ?

5

- 33.** (a) (i) What is a wavefront ? What is the direction of a beam of light with respect to the wavefront associated with it ?
- (ii) State Huygens' principle. Using it, explain the propagation of light.
- (iii) State two conditions necessary for light from two sources to be coherent.

5

OR

- (b) (i) Name the factors on which the focal length of a lens depends.
- (ii) A converging lens is immersed in a transparent medium. Explain the condition under which the lens (1) becomes invisible and (2) behaves as a diverging lens.
- (iii) Is it possible for two lenses in contact to have zero power ? Explain.

5



SECTION E

34. Case Study : A charged body inside an electric field.

A charged latex sphere of mass 1.85×10^{-13} kg is held stationary in between two horizontal plates which are separated by a distance 0.62 cm. The potential difference between the plates is 1.24×10^3 V with the upper plate being positive.

Based on the above facts, answer the following questions :

- (i) What is the nature of charge on the latex sphere ? 1
- (ii) What is the direction of electric field between the plates ? 1
- (iii) What is the magnitude of electric field between the plates ? 2

OR

- (iii) What is the magnitude of charge on the latex sphere ?
(Take $g = 10 \text{ ms}^{-2}$) 2

35. When a ray of light passes from a denser to a rarer medium, it bends away from the normal. If we go on increasing the angle of incidence in denser medium, then for a particular value of angle of incidence, the incident ray is refracted along the interface of the two media. This angle is called critical angle (i_c). If incident angle (i) is greater than critical angle (i_c), then the ray of light is reflected back in the same denser medium. This phenomenon is called total internal reflection of light.



Based on the above facts, answer the following questions :

- (i) If the refractive index of the denser medium increases, how will the corresponding value of critical angle for the pair of mediums change ? 1
- (ii) Explain how this phenomenon can be used in a prism to deviate a ray of light by 90° . 1
- (iii) What is an optical fibre ? Write its two uses. 2

OR

- (iii) A ray of light travels from an optically denser medium to a rarer medium and is incident at critical angle (i_c). Find the angle of deviation for this ray. 2